Sales Trend and Time-Based Performance Analysis for Afficionado Coffee Roasters

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Abstract**—***This study analyzes sales trends and time-based performance patterns for Afficionado Coffee Roasters using transaction-level data. The objective of the research is to identify temporal demand patterns and evaluate how sales performance varies across different time dimensions, including hours of the day, days of the week, and months. Understanding these patterns helps retail managers optimize staffing schedules, improve inventory planning, and enhance operational efficiency. The dataset includes transaction details such as transaction time, store location, product category, quantity sold, and revenue. Exploratory Data Analysis (EDA) techniques were applied using tools such as Excel, Python, and visualization libraries to examine overall sales trends and identify peak demand periods. Various analytical methods, including daily sales trend analysis, day-of-week performance evaluation, monthly sales aggregation, and hourly demand heatmaps, were used to extract meaningful insights from the data.*

The results indicate that sales show clear temporal patterns, with higher demand during morning and afternoon hours, particularly on weekends. Certain store locations also demonstrate stronger performance due to higher customer traffic. These findings highlight the importance of time-based sales analysis in retail operations. By leveraging these insights, businesses can improve workforce scheduling, adjust operating hours, and implement targeted promotional strategies to maximize revenue and customer satisfaction.

Overall, this research demonstrates how data-driven analysis of sales trends can support more effective decision-making in specialty coffee retail environments.

Index Terms*: — Sales Trend Analysis, Time-Based Performance, Retail Analytics, Coffee Shop Sales, Temporal Demand Patterns, Exploratory Data Analysis.*

1.INTRODUCTION

Retail businesses, particularly specialty coffee shops, rely heavily on timing of customer demand. Unlike many retail sectors, coffee purchases are highly dependent on daily routines such as morning commutes, lunch breaks, and afternoon work breaks. A lack of understanding of these demand patterns can lead to operational challenges such as:

- Overstaffing during slow hours
- Understaffing during peak demand
- Longer customer wait times
- Inefficient operational costs

Retail managers often rely on intuition-based decision making, such as assuming mornings are always busy. However, data-driven analysis provides quantitative evidence that enables more accurate operational planning.

This study analyzes transaction-level sales data from Afficionado Coffee Roasters to understand time-based demand patterns and provide actionable recommendations.

2.PROBLEM STATEMENT

Coffee retail businesses experience fluctuating sales across different times of the day, days of the week, and months. Without proper analysis, it becomes difficult to identify peak demand periods and low-sales intervals. Lack of time-based insights may lead to inefficient staff scheduling, inventory shortages, or overstocking. managers often rely on intuition instead of data-driven decision making when planning operations. There is a need to analyze historical transaction data to understand sales trends and temporal demand patterns. Identifying these patterns will help optimize store operations, improve customer service, and increase revenue performance.

3.RELATED WORK

Previous studies in retail analytics have focused on analyzing sales patterns using historical transaction data to understand customer purchasing behavior.

Research on time-series analysis in retail shows that sales often vary by hour of the day, day of the week, and seasonal trends. Many retail businesses use Exploratory Data Analysis (EDA) and visualization techniques to identify peak demand periods and customer traffic patterns. Studies on coffee shop and food service industries highlight that morning and evening hours typically generate higher sales due to commuter and leisure demand. Data-driven retail management models suggest that analyzing temporal sales patterns helps improve staff scheduling, inventory planning, and marketing strategies. Visualization tools such as heatmaps, trend charts, and dashboards are widely used to monitor and interpret sales performance over time.

4.RESEARCH OBJECTIVES

The main objectives of this research are:

- Analyze overall sales trends across time
- Evaluate day-of-week sales performance
- Identify hourly demand patterns
- Compare revenue across store locations
- Develop data-driven recommendations for operational improvements

5. DATASET DESCRIPTION

The dataset used in this study contains transaction-level data from Afficionado Coffee Roasters. Key variables include:

Dataset Features:

TABLE-I

VARIABLE	DESCRIPTION
transaction_id	Unique identifier per transaction
year	Transaction year(2025)
transaction_time	Time of transaction (HH:MM:SS)
transaction_qty	Quantity purchased
unit_price	Price per unit
store_id	Store identifier
store location	Physical store location
product_id	Unique identifier for each product
product_category	Broad product group
product_type	Product variant within the category
product_detail	Detailed attributes such as flavor or blend

From these fields, additional features were derived such as:

- **Hours of transaction**
- **Day of week**
- **Time_bucket**
- **revenue**

This structured dataset enables detailed analysis of sales trends, hourly demand patterns, store location performance, and product category contributions, supporting data-driven decision making for retail operations.

6. METHODOLOGY

The study analyzes sales data from Afficionado Coffee Roasters to identify time-based sales trends and performance patterns. The dataset contains transaction details such as date and time, revenue, and transaction quantity. Data preprocessing was performed to clean the dataset and convert the transaction date-time into a standard format. Additional features such as day, month, hour, and day of the week were extracted to support time-based analysis. The cleaned data was aggregated to evaluate sales performance at different temporal levels including daily, weekly, monthly, hourly, and day-of-week. Exploratory Data Analysis (EDA) techniques were applied to identify trends and patterns in customer purchasing behavior. An interactive dashboard was developed using the Streamlit framework, with visualizations created using Plotly. The dashboard allows users to analyze sales trends dynamically based on different metrics such as revenue and transaction quantity and also enables downloading analyzed data in CSV format. The methodology helps provide insights into peak sales periods and overall store performance.

TABLE-II
KEY PERFORMANCE INDICATORS

KPI	Descriptio n	Formula / Measurement
Total Revenue	Total income from coffee sales.	Quantity × Unit Price
Total Transaction s	Total number of customer purchases.	Count of transactions
Average Transaction Value	Average spending per transaction.	Revenue / Transactions
Daily Sales Trend	Sales variation by day.	Daily revenue/transaction s
Weekly Sales Performanc e	Sales comparison by weekday.	Revenue by day of week
Monthly Sales Growth	Sales change across months.	Monthly revenue
Hourly Demand Pattern	Sales distribution by hour.	Revenue by hour
Store Location Performanc e	Sales comparison by location.	Revenue by store

7.SYSTEM DESIGN AND IMPLEMENTATION

1. System Overview

The system is designed to analyze sales data from Afficionado Coffee Roasters to identify temporal sales trends and performance patterns. It processes transaction data and generates visual insights such as daily, weekly, and hourly sales trends. The system helps managers understand customer demand and improve operational decision-making.

2. Data Collection

The dataset contains transactional information such as:

- Transaction ID
- Transaction Time
- Store Location
- Product Category
- Transaction Quantity
- Unit Price

This data is used to calculate revenue and analyze sales patterns across different time dimensions.

3. Data Processing

Data preprocessing is performed using Python and Pandas. The steps include:

- Converting transaction time into datetime format
- Creating new time features such as hour, day of week, and month
- Calculating revenue (transaction quantity × unit price)
- Cleaning and organizing the dataset for analysis.

4. Data Visualization

Interactive visualizations are created using Streamlit, Plotly, and Matplotlib. The dashboard displays:

- Overall sales trends
- Daily and weekly sales performance
- Hourly demand patterns
- Store location comparison

These visualizations allow users to easily interpret sales performance.

5. Dashboard Implementation

The system is implemented as an interactive Streamlit dashboard where users can:

- Select store locations
- Filter data by day and hour
- Toggle between revenue and transaction quantity

The dashboard dynamically updates charts based on user selections, enabling real-time exploratory data analysis.

TABLE-III
Tools and Technologies

Tool	Purpose
Python	Data analysis
Pandas	Data manipulation
Matplotlib	Static visualizations
Plotly	Interactive visualizations
Streamlit	Dashboard development

Fig. 1. Main dashboard interface with overview cards and filter panel

8.RESULTS AND DISCUSSION

Exploratory Data Analysis (EDA)

8.1a Daily Sales Trend

The results of the analysis provide insights into the sales performance and customer demand patterns of Afficionado Coffee Roasters across different time periods. The study examined daily, weekly, monthly, hourly, and day-of-week sales trends using revenue and transaction quantity data. The daily sales trend shows that revenue generally ranges between ₹1,700 and ₹2,300, with most days recording around ₹1,850–₹2,000. Transaction quantities remain relatively stable, typically between 560 and 620 transactions per day, indicating consistent customer demand.

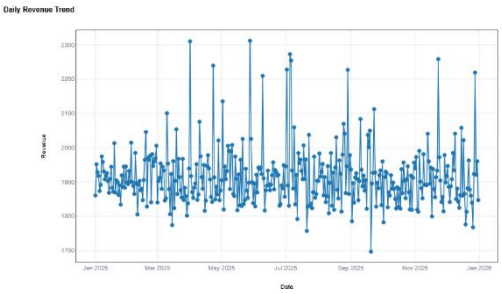

Fig. 2. Daily revenue Trend Analysis for Afficionado coffee Roasters.

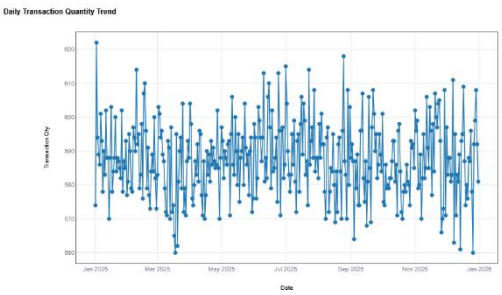

Fig. 2.a. Daily Transaction Quantity Trend Analysis for Afficionado coffee Roasters.

8.1b Weekly Sales Trend

The weekly sales trend indicates that weekly revenue mostly falls between ₹13,000 and ₹14,000, with transaction quantities around 4,000 transactions per week, showing stable performance across weeks.

Fig. 3. Weekly revenue Trend Analysis for Afficionado coffee Roasters.

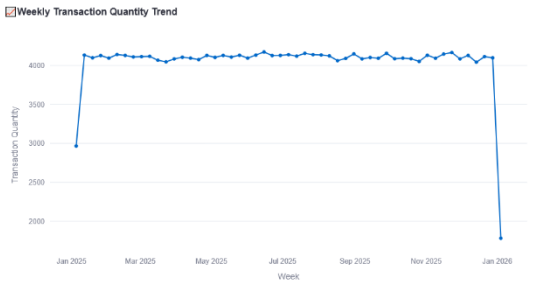

Fig. 3.a. Weekly Transaction Quantity Trend Analysis for Afficionado coffee Roasters.

8.1c Monthly Sales Trend

The monthly analysis reveals that total revenue ranges approximately between ₹53,000 and ₹60,000, while monthly transaction quantities vary from 16,000 to 18,000 transactions, suggesting steady long-term sales performance.

Fig. 4. Monthly revenue Trend Analysis for Afficionado coffee Roasters.

Fig. 4.a. Monthly Transaction Quantity Trend Analysis for Afficionado coffee Roasters.

8.2 Day-of-Week Performance

The day-of-week sales analysis shows slight variations in both transaction quantity and revenue across the week. The highest performance is observed on the day with 31,198 transactions and revenue of 102,377.2, indicating the peak demand period. This suggests that customer purchasing activity is strongest on this day, likely due to higher foot traffic or increased consumer spending.

Moderate performance is seen on days with 30,603 transactions (99,533.58 revenue) and 30,660 transactions (100,570.15 revenue), which also contribute significantly to total weekly sales. These values indicate consistent demand and stable revenue generation.

On the other hand, the lowest revenue occurs on the day with 30,529 transactions and revenue of 98,458.59, showing slightly reduced sales compared to other days. However, the variation across the week is relatively small, suggesting that coffee demand remains stable throughout the week with only minor fluctuations.

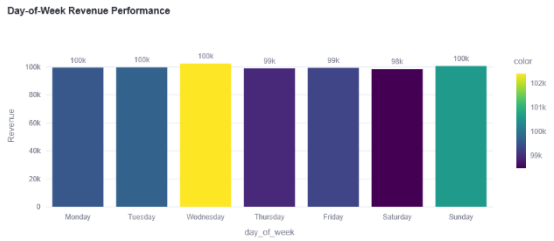

Fig. 5. Day of Week Revenue Performance of coffee Sales

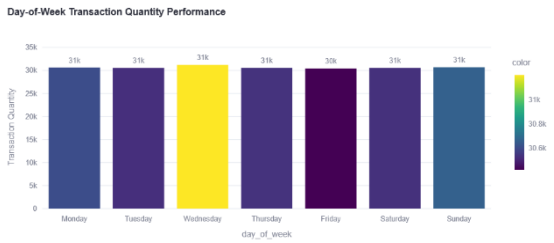

Fig. 5.a. Day of Week Transaction Quantity Performance of coffee Sales

8.3 Hourly Demand Patterns

The hourly sales data shows that coffee demand starts moderately at 6:00 AM and increases rapidly between 7:00 AM and 10:00 AM. **The** peak demand occurs around 9:00–10:00 AM, where revenue reaches about 12,000–13,000 and transactions are around 3600–3800**.**

After 10:00 AM, demand decreases, with revenue dropping to around 6300–7200 and transactions to about 1900–2000 by 11:00 AM**.** During the afternoon (12:00–17:00**)**, sales remain stable at moderate levels.

In the evening (18:00–20:00), both revenue and transactions decline significantly, indicating lower customer activity. Overall, the analysis shows that morning hours are the most important sales period for the coffee shop.

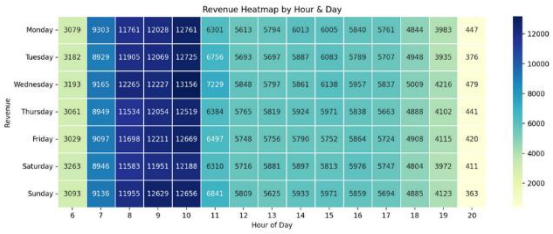

Fig. 6. Revenue of Hourly Demand Heatmap Showing sales distribution

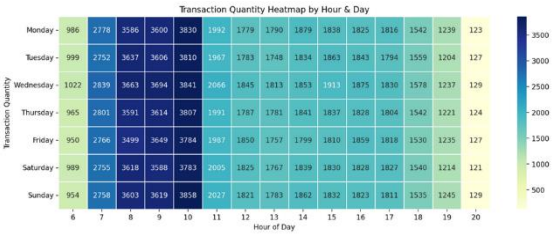

Fig. 6.a. Transaction of Hourly Demand Heatmap Showing sales distribution

8.4 Store Location Comparison

The provided data compares the revenue and transaction quantity across three store locations.

- Store 2 generated the highest revenue (236,511.17) with 71,737 transactions, indicating the strongest sales performance among the three locations.
- Store 1 recorded 232,243.91 revenue with 70,991 transactions, showing slightly lower performance compared to Store 2.
- Store 3 had the lowest revenue (230,057.25) despite having 71,742 transactions, which is the highest transaction count.

Discussion:
Although Store 3 processed the highest number of transactions, its revenue is the lowest, suggesting that customers may be purchasing lower-priced items or smaller orders. In contrast, store 2 achieves the highest revenue with slightly fewer transactions, indicating higher average sales per transaction. This suggests better product mix or higher-value purchases at that location.
Overall, store 2 demonstrates the best sales efficiency, while Store 3 may benefit from strategies such as upselling or promoting higher-value products to increase revenue

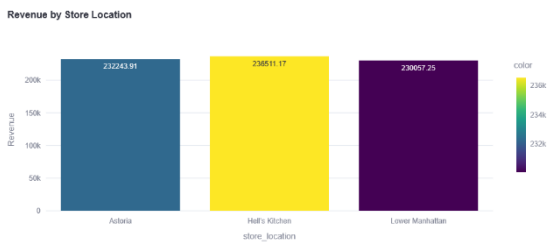

Fig. 7. Revenue comparison Across store location

Fig. 7.a. Transaction Quantity Comparison Across store location

9.LIMITATIONS

1. Limited Dataset Scope
The analysis depends on the available transaction dataset. If the data covers only a short time period or limited stores, the results may not fully represent long-term sales patterns.
2. Lack of External Factors
The analysis mainly focuses on time-based sales data and does not consider external factors such as weather conditions, local events, holidays, or economic changes, which can influence customer demand.
3. No Customer Behavior Insights
The dataset includes transactions and revenue but does not contain detailed customer demographic or preference information, limiting deeper behavioral analysis.
4. Product-Level Variations Not Fully Considered
Differences in product types, pricing strategies, or seasonal menu items may affect sales, but time-based analysis alone may not capture these product-specific impacts.
5. Store Operational Differences
Variations in staffing levels, service quality, store layout, or promotions across locations may influence sales performance but are not included in the dataset.
6. Data Quality Issues
Missing values, inconsistent timestamps, or recording errors in transaction data may affect the accuracy of the analysis.
7. Short-Term Trend Interpretation
Temporal analysis identifies patterns such as peak hours or busy days, but it may not accurately

predict long-term demand without additional forecasting methods.

8. Limited Predictive Capability

The study mainly focuses on descriptive analysis rather than advanced predictive models, which may limit the ability to forecast future sales trends.

10.ETHICAL NOTE

This study ensures that sales data is used responsibly and ethically. Customer privacy and confidentiality are maintained by avoiding the use of personal information. The analysis is conducted with accuracy, transparency, and fairness, ensuring that the results are presented honestly without data manipulation. The data is used strictly for research and analytical purp**oses** to support business insights and decision-making.

11.CONCLUSION

The Sales Trend and Time-Based Performance Analysis for Afficionado Coffee Roasters provides valuable insights into sales patterns across different time periods. The study identifies key trends in daily, weekly, and hourly sales performance, helping to understand customer purchasing behavior. The analysis highlights peak demand periods, store performance differences, and revenue patterns, which can support better operational planning.

Overall, the findings help coffee shop managers optimize staffing, improve inventory management, and enhance sales strategies. By using data-driven insights, businesses can make informed decisions to increase efficiency, improve customer service, and maximize revenue. Future enhancements such as predictive analytics and customer behavior anal**ysis** can further improve business performance and strategic planning.

12.FUTURE SCOPE

Future research can expand this study by analyzing larger and longer-term datasets to identify more accurate sales patterns. Advanced machine learning and forecasting models can be applied to predict future coffee demand. Additionally, incorporating customer behavior, product-level analysis, and external factors such as weather and holidays can provide deeper insights. Developing real-time analytical dashboards can also help store managers make faster and more effective business decisions.

REFERENCES

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